Freight Rate Prediction — Validation Approach & December Forecast

Machine Learning Engineer Assessment · Model: CatBoost (selected over XGBoost)

1. Train / Test Validation Approach

Split methodology

The labeled development data (train-test.csv, 48,000 rows) was split chronologically into an 80% training set and a 20% held-out test set, using date as the split key rather than a random sample. This choice reflects the nature of the task: validation.csv covers November-December 2025, a period entirely after the training data's range (January-October 2025) — meaning the real deployment scenario is forecasting into an unseen future period, not filling in random gaps within an already-observed period. Sorting by date and holding out the most recent 20% (2025-08-31 to 2025-10-31) as the test set mirrors that scenario directly: the model is evaluated on data that chronologically follows its training window, exactly as it will be when scored against November-December loads. A random split was considered but rejected, since it would let the test set contain rows interleaved in time with the training data — overstating accuracy relative to true forecasting performance by letting the model implicitly interpolate within a period it has already seen.

Data-quality checks performed before splitting

- **Negative weight values** (292 rows, ~0.6%): physically impossible; treated as a sign error and corrected with an absolute-value transform rather than dropped, since the magnitude of the values looked plausible once corrected.
- **Missing values** in weight (300 rows) and market_index (374 rows): left as NaN — both XGBoost and CatBoost natively handle missing values by learning an optimal split direction for them, so no imputation was required.
- **Outlier audit across all numeric columns** using the IQR method, followed by manual verification of every flagged column. Longitude and quote_signal were flagged by IQR but confirmed to be naturally skewed, valid distributions (not errors) upon inspection. Distance and posted_rate outliers were confirmed to be legitimate long-haul / high-value loads, not data errors.
- **Target skew**: posted_rate is right-skewed — 255 loads (~0.5%) exceed $6,457 (the IQR upper bound), some as high as $17,900. These are longer-haul, disproportionately Reefer/Flatbed loads with a rate-per-mile roughly 3x the norm — a legitimate rare market segment, not noise. The model was trained on log1p(posted_rate) to reduce the influence of this skew on the loss function, with predictions inverse-transformed (expm1) before every evaluation and output step.

Feature engineering

date was decomposed into day_of_week, month, day_of_year, and year to expose seasonal and weekly patterns to the model (freight demand varies by both season and day of week). pickup, delivery, and equipment were treated as native categorical features — pandas 'category' dtype for XGBoost, string dtype with CatBoost's built-in ordered target statistics for CatBoost — rather than one-hot encoded, since both models handle categorical splits natively and this avoids the dimensionality blow-up from one-hot encoding high-cardinality city columns.

Model comparison on the held-out (chronological) test set

Test period: 2025-08-31 to 2025-10-31 (the most recent 20% of the training window by date).

Model	MAE ($)	RMSE ($)	R-squared	MAPE
XGBoost	168.31	661.47	0.8119	7.62%
CatBoost (selected)	114.11	635.21	0.8265	4.97%

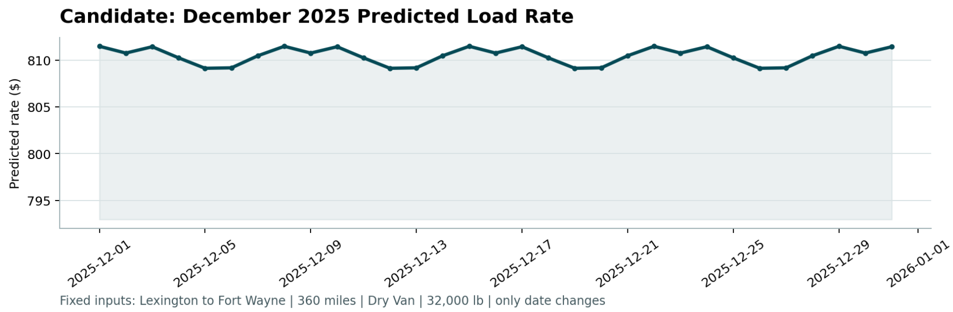
CatBoost was selected as the final model (lower MAE, RMSE, MAPE and higher R-squared across the board). The gap is largely attributable to CatBoost's ordered target-statistics encoding, which handles the high-cardinality pickup/delivery city columns more effectively than XGBoost's native categorical splits. A residual analysis on the time-based test set confirmed both models' largest errors overlap completely (20/20 of the worst-predicted test rows were shared between the two models) and correspond entirely to the rare, high-value load segment (posted_rate > $6,457 — 61 of 9,600 test rows, ~0.6%) described above — a data characteristic both models struggle with equally, rather than a model-specific weakness.

Validation-set inference

The selected CatBoost model — trained once on train-test.csv — was applied via .predict() to all 12,000 rows of validation.csv to produce validation_predictions.csv (load_id, predicted_rate). 1,447 of those rows (12.1%) reference pickup/delivery cities not present in the training data; these are flagged internally as lower-confidence, since CatBoost falls back to feature-based estimation rather than a learned city-specific effect for unseen categories.

2. Fixed December 2025 Prediction Chart

The same trained CatBoost model (no retraining) was used to predict rates for a fixed load — Lexington to Fort Wayne, Dry Van, 360 miles, 32,000 lb — across every day of December 2025, with only the date changing. Route-specific proxy values (median market_index and quote_signal, and exact pickup/delivery coordinates) were pulled from the 21 historical Lexington→Fort Wayne Dry Van loads in the training data, since the December template does not include these columns. This chart was produced by the provided scorer.py script from data/december_chart_inputs.csv after the predicted_rate column was filled in.



The predicted rate follows a clear 7-day cycle — modestly higher on weekdays and lower around weekends — consistent with typical freight demand patterns, indicating the model has learned a sensible day-of-week effect rather than an arbitrary one.